

CHAPTER 1 – PROJECT OVERVIEW

1.1 Introduction

The **Smart Asset Management System (SAMS)** is a web-based enterprise application developed to manage organizational assets throughout their lifecycle. The system centralizes the management of departments, employees, vendors, asset categories, physical assets, asset assignments, maintenance requests, and damage reports through a secure role-based interface.

The primary objective of this application is to replace manual asset tracking with a digital system that improves inventory visibility, enhances accountability, and streamlines operational workflows.

The application follows a layered architecture using Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Thymeleaf, and PostgreSQL, making it scalable, maintainable, and aligned with modern enterprise development practices.

1.2 Objectives

The primary objectives of the Smart Asset Management System are:

- Centralize asset information in a single system.
 - Track the complete lifecycle of organizational assets.
 - Assign assets to employees efficiently.
 - Maintain accurate vendor and procurement information.
 - Record maintenance history and damage reports.
 - Enforce business rules through application-level validations.
 - Provide role-based access to system users.
 - Generate live dashboard statistics for administrative decision-making.
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1.3 Project Scope

The current implementation supports:

- Secure login using database authentication
- Role-based authorization
- Department Management
- Employee Management
- Vendor Management
- Asset Category Management
- Asset Inventory Management
- Asset Assignment

- Maintenance Management
- Damage Reporting
- Dashboard Analytics
- Swagger API Documentation

Future versions may include:

- QR Code Integration
- Email Notifications
- Excel Export
- PDF Reports
- Docker Deployment
- CI/CD Pipeline
- Cloud Deployment

CHAPTER 2 – SYSTEM ARCHITECTURE

2.1 Architecture Overview

The application follows a layered architecture.

Presentation Layer

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Spring MVC Controllers

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Service Layer

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Repository Layer

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PostgreSQL Database

Each layer has a specific responsibility:

Layer	Responsibility
Presentation	Handles user requests and UI rendering
Controller	Accepts HTTP requests
Service	Implements business logic
Repository	Communicates with PostgreSQL
Database	Stores application data

2.2 Technology Stack

Category	Technology
Programming Language	Java 21
Backend Framework	Spring Boot
MVC Framework	Spring MVC
Security	Spring Security
ORM	Hibernate
Data Access	Spring Data JPA
Database	PostgreSQL
Frontend	Thymeleaf
Styling	Bootstrap 5
Icons	Bootstrap Icons
Charts	Chart.js
Build Tool	Maven
Documentation	Swagger OpenAPI
IDE	Spring Tool Suite (STS)

CHAPTER 3 – LOGIN & SECURITY MODULE

3.1 Module Overview

The Login Module authenticates users before allowing access to the system. Authentication is performed using credentials stored in the PostgreSQL database.

Passwords are encrypted using the BCrypt hashing algorithm before being stored.

After successful authentication, users are redirected according to their assigned role.

3.2 User Roles

The application currently supports two user roles.

Administrator

The Administrator has complete access to all modules.

Accessible Modules:

- Dashboard
 - Department Management
 - Employee Management
 - Vendor Management
 - Asset Categories
 - Asset Management
 - Asset Assignment
 - Maintenance
 - Damage Reports
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Employee

The Employee has restricted access.

Current Access:

- Employee Dashboard

The employee module serves as the foundation for future enhancements such as viewing assigned assets, submitting damage reports, and tracking maintenance requests.

3.3 Authentication Workflow

User opens Login Page

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Enter Username & Password

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Spring Security Authentication

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Database Verification

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Password Verification (BCrypt)

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Authentication Successful

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Role Check

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ADMIN

EMPLOYEE

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Dashboard Employee Dashboard

3.4 Validations

The login module performs the following validations:

- Username must exist.
- Password must match the encrypted password stored in the database.

- Unauthorized users cannot access protected URLs.
 - Users without the required role receive an Access Denied page.
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3.5 Security Features

The security module implements the following:

- Spring Security
 - Database Authentication
 - BCrypt Password Encryption
 - Session Management
 - Role-Based Authorization
 - Custom Login Page
 - Access Denied Handling
 - URL Protection
 - Logout Functionality
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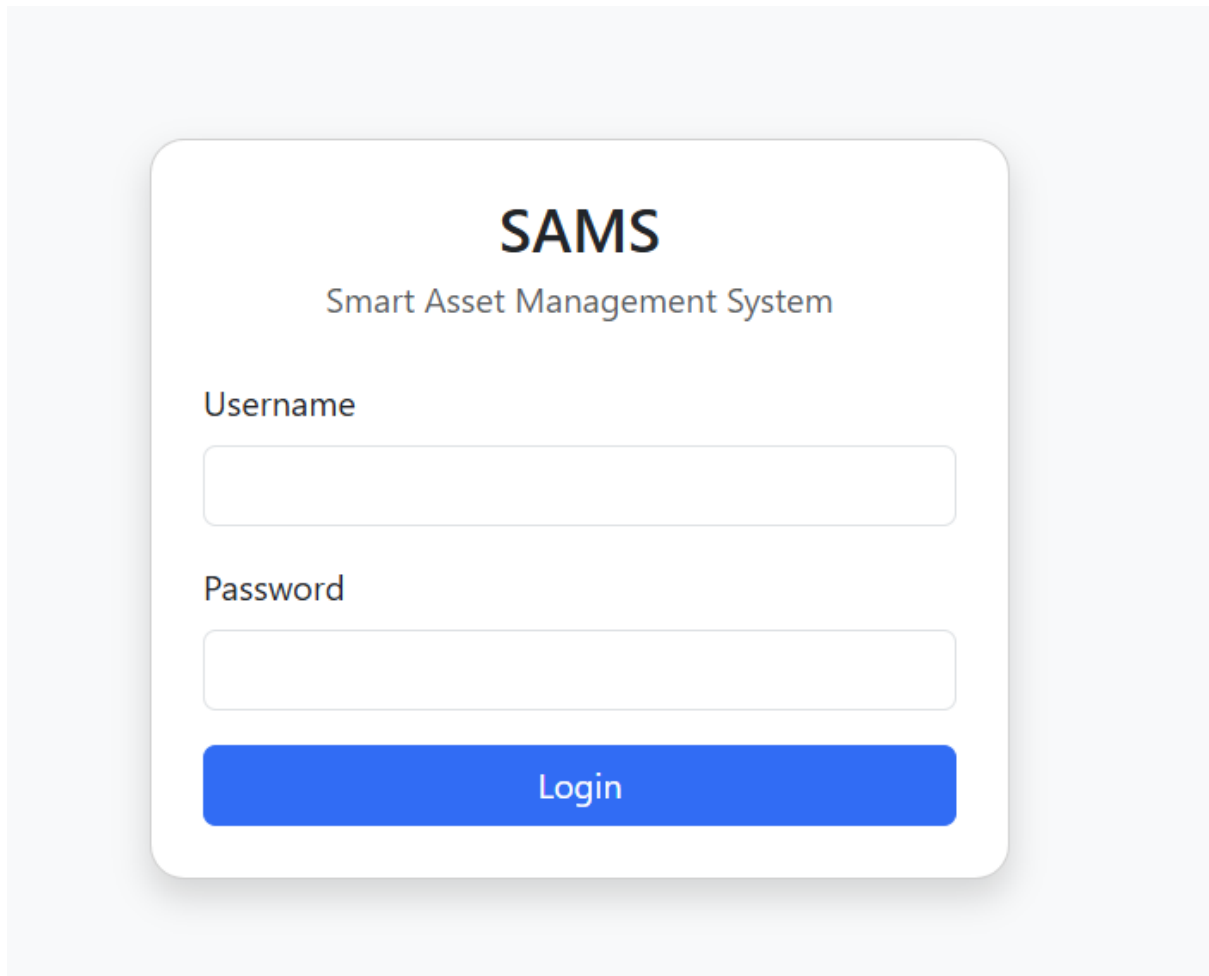
3.6 Edge Cases

The system properly handles the following situations:

Scenario	System Response
Invalid username	Login failed
Incorrect password	Login failed
Employee accessing Admin URL	Access Denied
Direct URL access without login	Redirect to Login
Session expired	Redirect to Login
Unauthorized module access	Access Denied Page

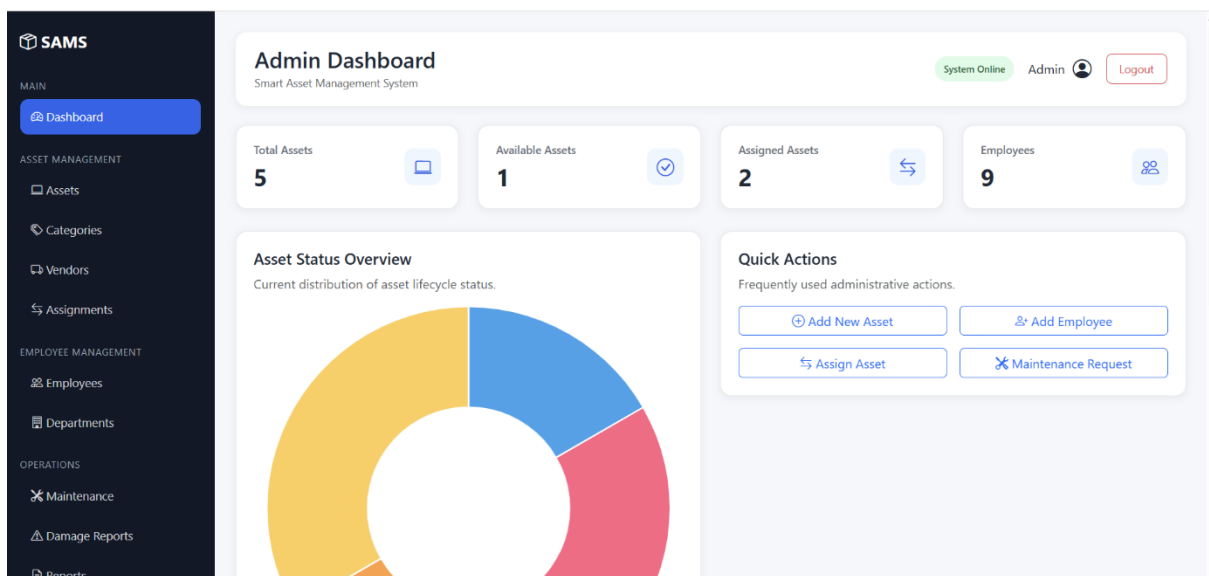
3.7 Screenshots

Figure 3.1 – Login Page



The screenshot shows the login page for the Smart Asset Management System (SAMS). The page has a light gray background. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle, the text "SAMS" is displayed in a large, bold, black font, followed by "Smart Asset Management System" in a smaller, regular black font. Below this, there are two input fields: "Username" and "Password", each with a light gray border and a small icon on the left. A blue button with the text "Login" in white is positioned below the password field.

Figure 3.2 – Administrator Dashboard after Login



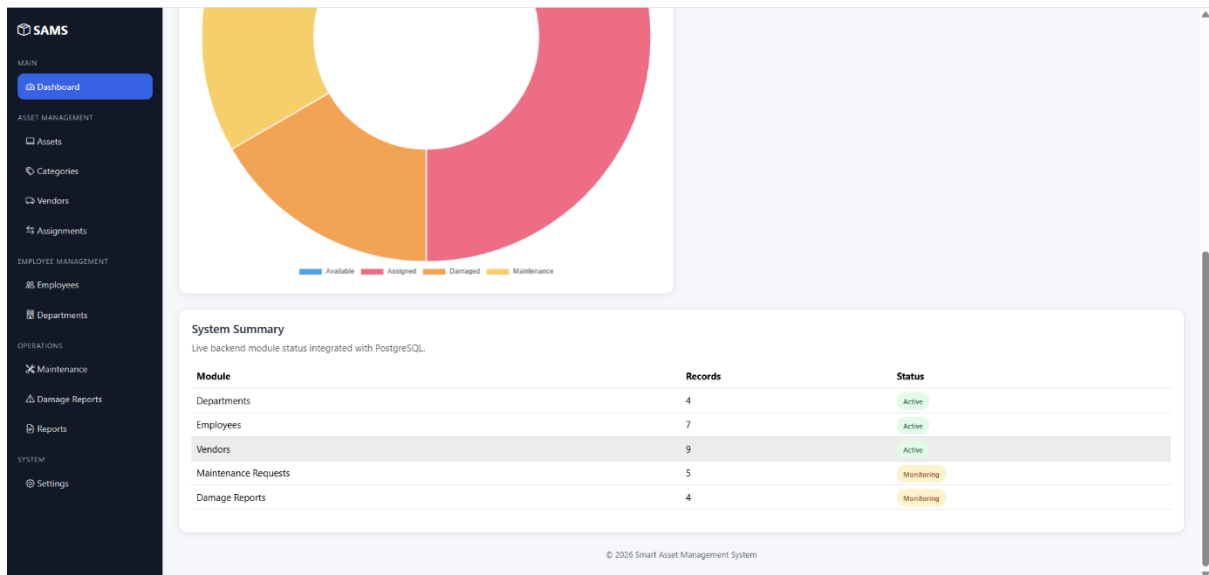
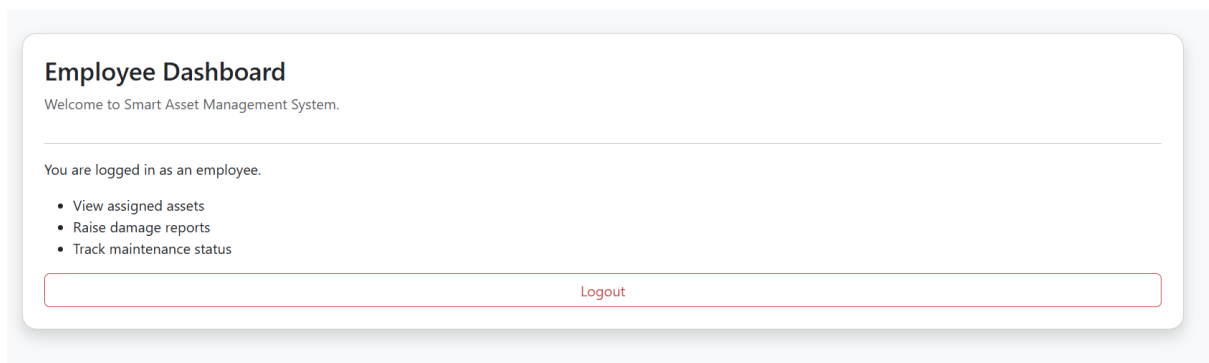


Figure 3.3 – Employee Dashboard after Login(For Now)



Department Management Module

The Department module is responsible for managing organizational departments. It serves as the foundation for employee organization, as every employee is associated with a department.

Features

- Add new department
- View all departments
- Update department information
- Delete department
- Automatically maintain created and updated timestamps

Validations

- Department name is mandatory.
- Department description is mandatory.
- Department name cannot exceed 100 characters.

- Description cannot exceed 250 characters.
- Duplicate department names are not allowed.

Business Rules / Edge Cases

- A department linked with employees cannot be deleted.
- New departments are assigned **ACTIVE** status by default.
- Invalid department IDs return an appropriate error message.
- Empty or whitespace-only department names are rejected.

SAMS
Smart Asset Management System

System Online Admin Logout

Departments

Manage organization departments

[Add Department](#)

ID	Department Name	Description	Status	Created At	Actions
3	HR Department	Handles employee records and HR operations	ACTIVE	2026-07-08T18:35:32.432863	Edit Delete
1	Human Resource Department	Handles HR operations and employee records	ACTIVE	2026-07-08T00:00:55.931523	Edit Delete
4	IT Information	This is a information Department	ACTIVE	2026-07-09T00:35:55.815172	Edit Delete
5	IT Department	this is a it department	ACTIVE	2026-07-09T00:41:17.499406	Edit Delete
6	Finance Department	Finance Department	ACTIVE	2026-07-10T11:12:33.967767	Edit Delete

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Employee Management Module

The Employee Management module maintains employee records within the organization. Each employee is mapped to a department and can later be assigned organizational assets.

Features

- Add employee
- View employee list
- Update employee details
- Delete employee
- Assign employee to a department
- Employee status management

Validations

- Employee Code is mandatory and must be unique.
- Employee name is mandatory and accepts alphabets only.

- Email address must be valid and unique.
- Phone number must contain exactly **10 digits**.
- Joining date cannot be a future date.
- Department selection is mandatory.

Business Rules / Edge Cases

- Duplicate employee codes are not allowed.
- Duplicate email addresses are not allowed.
- Invalid employee IDs return an appropriate error.
- Employee records maintain automatic created and updated timestamps.

SAMS

MAIN

Dashboard

ASSET MANAGEMENT

Assets

Categories

Vendors

Assignments

EMPLOYEE MANAGEMENT

Employees

Departments

OPERATIONS

Maintenance

Damage Reports

Reports

Admin Dashboard

Smart Asset Management System

System Online Admin Logout

Employees

Manage employee records and department mapping

Add Employee

ID	Code	Full Name	Email	Phone	Designation	Department	Status	Actions
2	EM02	Rajan Sharma	rajan@gmail.com	1236547890	Developer	HR Department	ACTIVE	Edit Delete
3	EM03	Hanna Sharma	hanna@gmail.com	9874563210	Developer	Human Resource Department	ACTIVE	Edit Delete
4	EM04	Nova Sharma	innova@gmail.com	1478523690	Programmer	IT Department	ACTIVE	Edit Delete
5	EM05	Nabisha	nabisha@gmail.com	3698521470	Tester	IT Information	ACTIVE	Edit Delete
6	EM10	Md Parvezuddin	mdparvez5546@gmail	7217742987	Developer	Human Resource Department	ACTIVE	Edit Delete
8	EM11	Parvezuddin	mdparvez5546@gmail.com	5217742987	tester	IT Information	ACTIVE	Edit Delete
9	EMP	Sunidhi	sunidhi89@gmail.com	8099080987	Programmer	IT Department	ACTIVE	Edit Delete
10	EMP01	jj	parvezjan2000@gmail.com	8099009877	Programmer	IT Department	ACTIVE	Edit Delete
1	EM01	Rahul Sharma	rahul.sharma@example.com	9876543210	Software Developer	Human Resource Department		Edit Delete

Vendor Management Module

The Vendor Management module stores information about vendors who supply organizational assets. Each asset is associated with a vendor for procurement and maintenance tracking.

Features

- Add vendor
- View vendor list
- Update vendor details
- Delete vendor
- Vendor status management

Validations

- Vendor name is mandatory.
- Contact person is mandatory.
- Email address must be in valid format.
- Phone number must contain exactly **10 digits**.
- Duplicate vendor names are not allowed.
- Duplicate email addresses are not allowed.

Business Rules / Edge Cases

- Vendors linked with existing assets cannot be deleted.
- Invalid vendor IDs return an appropriate error.
- Automatic timestamp management for created and updated records.

The screenshot displays the SAMS Admin Dashboard. On the left is a dark sidebar with the SAMS logo and a navigation menu including: MAIN (Dashboard), ASSET MANAGEMENT (Assets, Categories, Vendors, Assignments), EMPLOYEE MANAGEMENT (Employees, Departments), OPERATIONS (Maintenance, Damage Reports, Reports), and a Logout button. The main content area is titled 'Admin Dashboard' with the subtitle 'Smart Asset Management System'. It shows system status as 'System Online', the user as 'Admin', and a 'Logout' button. The 'Vendors' section is active, with a subtitle 'Manage asset suppliers and service partners' and an 'Add Vendor' button. A table lists 9 vendors with columns for ID, Vendor Name, Contact Person, Email, Phone, Address, Status, and Actions. The table data is as follows:

ID	Vendor Name	Contact Person	Email	Phone	Address	Status	Actions
1	Dell Technologies	Amit Verma	amit.verma@dellvendor.com	9876543210	Bangalore	ACTIVE	[Edit] [Delete]
2	Asus	Raman Kumar	ramandcd@g	44645	gcfvbn	INACTIVE	[Edit] [Delete]
3	HP	Jitesh	jitesh@gmail.com	8523697410	H-12 Srilankan Road	ACTIVE	[Edit] [Delete]
4	Samsung	Salim	salim@gmail.com	1236547890	Lovely professional university BH -5 907	ACTIVE	[Edit] [Delete]
5	Xiomi	Xeniii	xen@gmail.com	8529637410	H-18 Srilankan Road	ACTIVE	[Edit] [Delete]
6	Vivo	Parvez	cgvhb@g.nijk	9465214645	exrctvybuni	ACTIVE	[Edit] [Delete]
7	google	Md Parvezuddin	mdparvez5546@ail.co	7217742987	Punjab	ACTIVE	[Edit] [Delete]
8	Microsoft	Mohan	vvvhv@gmail.com	7485120854	erxtyuhi	ACTIVE	[Edit] [Delete]
9	Lava	Lala	xerty@gmail.com	7865412302	dfghbjnkml	ACTIVE	[Edit] [Delete]

Asset Category Management Module

The Asset Category module classifies assets into different categories, making inventory organization and reporting more efficient.

- **Features**
- Add asset category
- View category list
- Update category details
- Delete category
- Category status management

- **Validations**
- Category name is mandatory.
- Description is mandatory.
- Duplicate category names are not allowed.
- **Business Rules / Edge Cases**
- Categories associated with assets cannot be deleted.
- Invalid category IDs return an appropriate error.
- Newly created categories are assigned **ACTIVE** status by default.

The screenshot displays the SAMS Admin Dashboard. On the left is a dark sidebar with the SAMS logo and a menu including: MAIN (Dashboard), ASSET MANAGEMENT (Assets, Categories, Vendors, Assignments), EMPLOYEE MANAGEMENT (Employees, Departments), and OPERATIONS (Maintenance, Damage Reports, Reports). The 'Categories' link is highlighted. The main content area is titled 'Admin Dashboard' with the subtitle 'Smart Asset Management System'. It shows system status as 'System Online', the user as 'Admin', and a 'Logout' button. Below this is the 'Asset Categories' section with the subtitle 'Manage asset classification like laptop, desktop, monitor' and an 'Add Category' button. A table lists four categories: ID 1 (Laptop), ID 9 (Peripherals Devices), ID 8 (Phone), and ID 10 (Tablet). All are 'ACTIVE' and have edit/delete icons in the 'Actions' column. The footer shows '© 2026 Smart Asset Management System'.

ID	Category Name	Description	Status	Created At	Actions
1	Laptop	Portable computer assigned to employees	ACTIVE	2026-07-08T00:01:40.533590	Edit Delete
9	Peripherals Devices	This is the combination of keyboard, mouse, cables, adapter etc.	ACTIVE	2026-07-09T00:50:53.576269	Edit Delete
8	Phone	This is for the smartphones provided by the company	ACTIVE	2026-07-08T20:15:32.434988	Edit Delete
10	Tablet	Tablets	ACTIVE	2026-07-10T11:11:29.245264	Edit Delete

Asset Management Module

The Asset Management module is the core component of the Smart Asset Management System. It maintains complete information about organizational assets, including procurement details, vendor information, category mapping, warranty details, and asset lifecycle status.

Features

- Register new asset
- View all assets
- Update asset information
- Delete asset
- Map asset with category
- Map asset with vendor

- Track asset status
- Track asset condition

Validations

- Asset Code is mandatory and unique.
- Asset Name is mandatory.
- Serial Number is mandatory and unique.
- Purchase Cost must be greater than zero.
- Purchase Date cannot be a future date.
- Warranty Expiry Date must be after Purchase Date.
- Category selection is mandatory.
- Vendor selection is mandatory.

Business Rules / Edge Cases

- Duplicate Asset Codes are not allowed.
- Duplicate Serial Numbers are not allowed.
- Assets linked with assignments, maintenance requests, or damage reports cannot be deleted.
- Asset status is automatically set to **AVAILABLE** during registration.
- Asset condition is automatically set to **GOOD** during registration.

SAMS

MAIN

Dashboard

ASSET MANAGEMENT

Assets

Categories

Vendors

Assignments

EMPLOYEE MANAGEMENT

Employees

Departments

OPERATIONS

Maintenance

Damage Reports

Reports

Admin Dashboard

Smart Asset Management System

System Online Admin Logout

Assets

Manage company assets and inventory

Add Asset

ID	Asset Code	Name	Brand	Model	Category	Vendor	Status	Condition	Actions
4	LAP-002	HP latitude 090	HP	Latitude 6552	Laptop	HP	UNDER_MAINTENANCE	NEEDS_REPAIR	Edit Delete
3	LAP-003	74	Dell	Latitude 5440	Laptop	Dell Technologies	AVAILABLE	GOOD	Edit Delete
6	PHN-006	Asus latitude 96	Dell	74563	Phone	Dell Technologies	UNDER_MAINTENANCE	NEEDS_REPAIR	Edit Delete
5	PHN-005	Asus latitude 88	Asus	Latitude 852	Phone	Asus	ASSIGNED	GOOD	Edit Delete
8	PHN-009	Asus latitude 96	Asus	Latitude 5440	Phone	Microsoft	ASSIGNED	DAMAGED	Edit Delete

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Asset Assignment Module

The Asset Assignment module manages the allocation of company assets to employees. It records assignment details, return information, and ensures that assets are allocated according to predefined business rules.

Features

- Assign asset to employee
- View assignment history
- Return assigned asset
- Track assignment status
- Automatic asset status update

Validations

- Asset selection is mandatory.
- Employee selection is mandatory.
- Assignment date is automatically generated.
- Asset must exist before assignment.
- Employee must exist before assignment.

Business Rules / Edge Cases

- Only **AVAILABLE** assets can be assigned.
- Assets already assigned cannot be assigned again.
- Returned assets cannot be returned multiple times.
- Returning an asset automatically changes its status back to **AVAILABLE**.
- Invalid asset or employee IDs generate appropriate error messages.

The screenshot displays the SAMS Admin Dashboard. On the left is a dark sidebar with a menu including: MAIN (Dashboard), ASSET MANAGEMENT (Assets, Categories, Vendors, Assignments - highlighted), EMPLOYEE MANAGEMENT (Employees, Departments), and OPERATIONS (Maintenance, Damage Reports, Reports). The main content area is titled 'Admin Dashboard' with a subtitle 'Smart Asset Management System'. It includes a status bar with 'System Online', 'Admin' (with a profile icon), and a 'Logout' button. Below this is the 'Asset Assignments' section, which has a subtitle 'Track assigned and returned assets' and an 'Assign Asset' button. A table lists three assignments:

ID	Asset	Employee	Assigned Date	Expected Return	Actual Return	Status	Remarks	Actions
2	74	Hanna Sharma	2026-07-09	2026-07-23	Pending	ASSIGNED	You will be getting this in next two day.	[Edit] [Delete]
3	Asus latitude 88	Nabisha	2026-07-09	2026-07-09	Pending	ASSIGNED	You have to return on time	[Edit] [Delete]
4	Asus latitude 96	Parvezuddin	2026-07-10	2026-07-24	Pending	ASSIGNED		[Edit] [Delete]

At the bottom of the dashboard, there is a copyright notice: '© 2026 Smart Asset Management System'.

Maintenance Management Module

The Maintenance module records and manages maintenance activities for organizational assets. It tracks maintenance requests, service vendors, repair costs, and maintenance status throughout the repair lifecycle.

Features

- Create maintenance request
- View maintenance records
- Complete maintenance request
- Track maintenance status
- Associate maintenance with vendor

Validations

- Asset selection is mandatory.
- Vendor selection is mandatory.
- Issue description is mandatory.
- Maintenance cost cannot be negative.

Business Rules / Edge Cases

- Assigned assets cannot be sent for maintenance.
- Assets already under maintenance cannot have another active maintenance request.
- Completing maintenance automatically changes asset status to **AVAILABLE**.
- Asset condition is restored to **GOOD** after successful maintenance.
- Completed maintenance requests cannot be completed again.

SAMS

MAIN

Dashboard

ASSET MANAGEMENT

Assets

Categories

Vendors

Assignments

EMPLOYEE MANAGEMENT

Employees

Departments

OPERATIONS

Maintenance

Damage Reports

Reports

Admin Dashboard

Smart Asset Management System

System Online Admin Logout

Maintenance Requests

Track assets under repair and service lifecycle

New Maintenance

ID	Asset	Vendor	Issue	Cost	Start Date	End Date	Status	Remarks	Actions
1	74	Asus	breaking	6.0	2026-07-08	2026-07-08	COMPLETED	fix	
3	74	HP	Half screen is black	12000.0	2026-07-09	Pending	IN_PROGRESS	I want it next 3 days	 
4	HP latitude 090	Asus	Keys are not working properly	1400.0	2026-07-09	Pending	IN_PROGRESS	all keys are not working	 
2	74	Dell Technologies	gyg	3.0	2026-07-08	2026-07-09	COMPLETED	mm m	
5	Asus latitude 96	Samsung	wexrctvbn	20000.0	2026-07-10	Pending	IN_PROGRESS	xrcvbjnkml	 

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Damage Report Module

The Damage Report module enables administrators to record damaged assets, track their status, and maintain a history of damage reports. It helps ensure that damaged assets are identified, repaired, and returned to operational use.

Features

- Report damaged asset
- View damage reports
- Update report status
- Resolve damage report
- Record administrator remarks

Validations

- Asset selection is mandatory.
- Employee selection is mandatory.
- Damage description is mandatory.
- Report status must be selected during resolution.

Business Rules / Edge Cases

- Duplicate active damage reports for the same asset are not allowed.
- Assets under maintenance cannot be reported as damaged again.
- Resolving a damage report automatically updates the asset condition to **GOOD**.
- Resolved reports cannot be resolved multiple times.
- Invalid asset or employee IDs return appropriate error messages.

The screenshot displays the SAMS Admin Dashboard. On the left is a dark sidebar with a menu including: MAIN (Dashboard), ASSET MANAGEMENT (Assets, Categories, Vendors, Assignments), EMPLOYEE MANAGEMENT (Employees, Departments), and OPERATIONS (Maintenance, **Damage Reports**, Reports). The main content area is titled 'Admin Dashboard' with a subtitle 'Smart Asset Management System'. It shows system status as 'System Online', the user as 'Admin', and a 'Logout' button. The 'Damage Reports' section has a subtitle 'Track damaged assets and resolution status' and a 'Report Damage' button. Below is a table with 8 columns: ID, Asset, Employee, Description, Damage Date, Status, Admin Remarks, and Actions. The table contains 4 rows of data. The 'Status' column uses color-coded labels: PENDING (yellow) and RESOLVED (green). Each row has two action icons: a green checkmark and a red trash can.

ID	Asset	Employee	Description	Damage Date	Status	Admin Remarks	Actions
2	Asus latitude 88	Nova Sharma	Getting Shutdown Automatically	2026-07-09	PENDING	N/A	
1	74	Rahul Sharma	hgvhvh	2026-07-08	PENDING		
3	Asus latitude 88	Hanna Sharma	ezxrtcyvbn	2026-07-10	RESOLVED		
4	Asus latitude 96	Hanna Sharma	wzxrcybn	2026-07-10	PENDING	N/A	

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